



UE22CS352B - Object Oriented Analysis & Design

Mini Project Report

Movie Booking System

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FACULTY OF ENGINEERING
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Problem Statement:

The Movie Booking System is a web-based application designed to simplify the process of booking movie tickets online. The platform enables users to browse upcoming movie events and check seat availability in real-time, reserve seats, order food combos, and make payments using multiple payment gateways. Users can also access their booking history and cancel bookings. This system ensures a seamless and engaging user experience, eliminating the need to stand in queues at theaters or event counters.

Key Features:

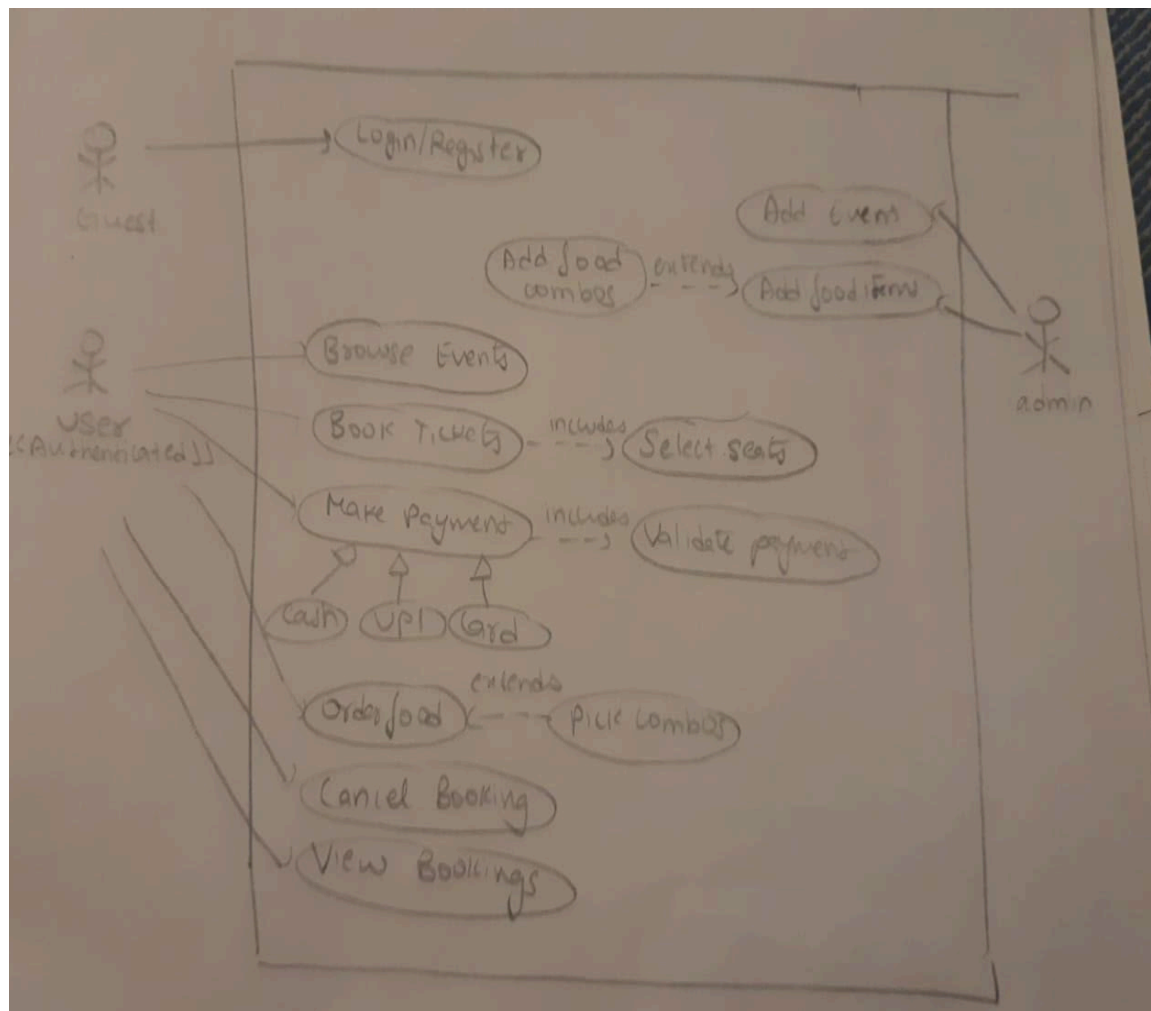
The system offers a wide range of features that enhance the booking experience for users and simplify event management for administrators:

- **User Authentication:** Secure login and signup functionality with role-based access.
- **Event Catalog:** Dynamic listing of available movies and events with details.
- **Real-Time Seat Selection:** Interactive seat map showing real-time availability and allowing users to reserve their preferred seats.
- **Food Combo Ordering:** Users can select and order snacks and combo offers during the booking process.
- **Payment Gateway Integration:** Multiple payment methods supported including CARD, UPI, and CASH.
- **Invoice Generation:** Users can view and download their invoice with booking, food order and billing information.

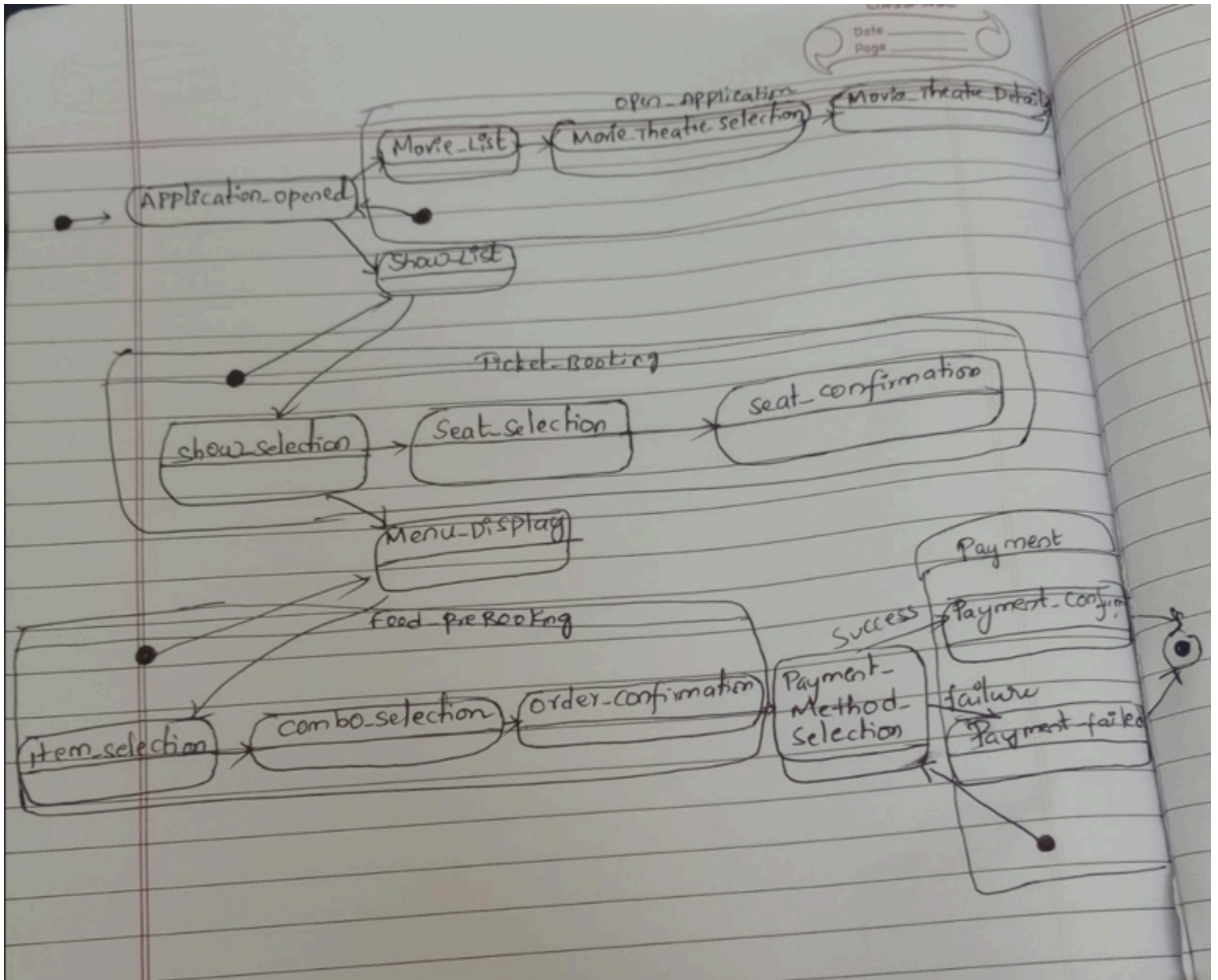
Models:

- User
- Event
- Booking
- Food Order
- Food Combo
- Payment

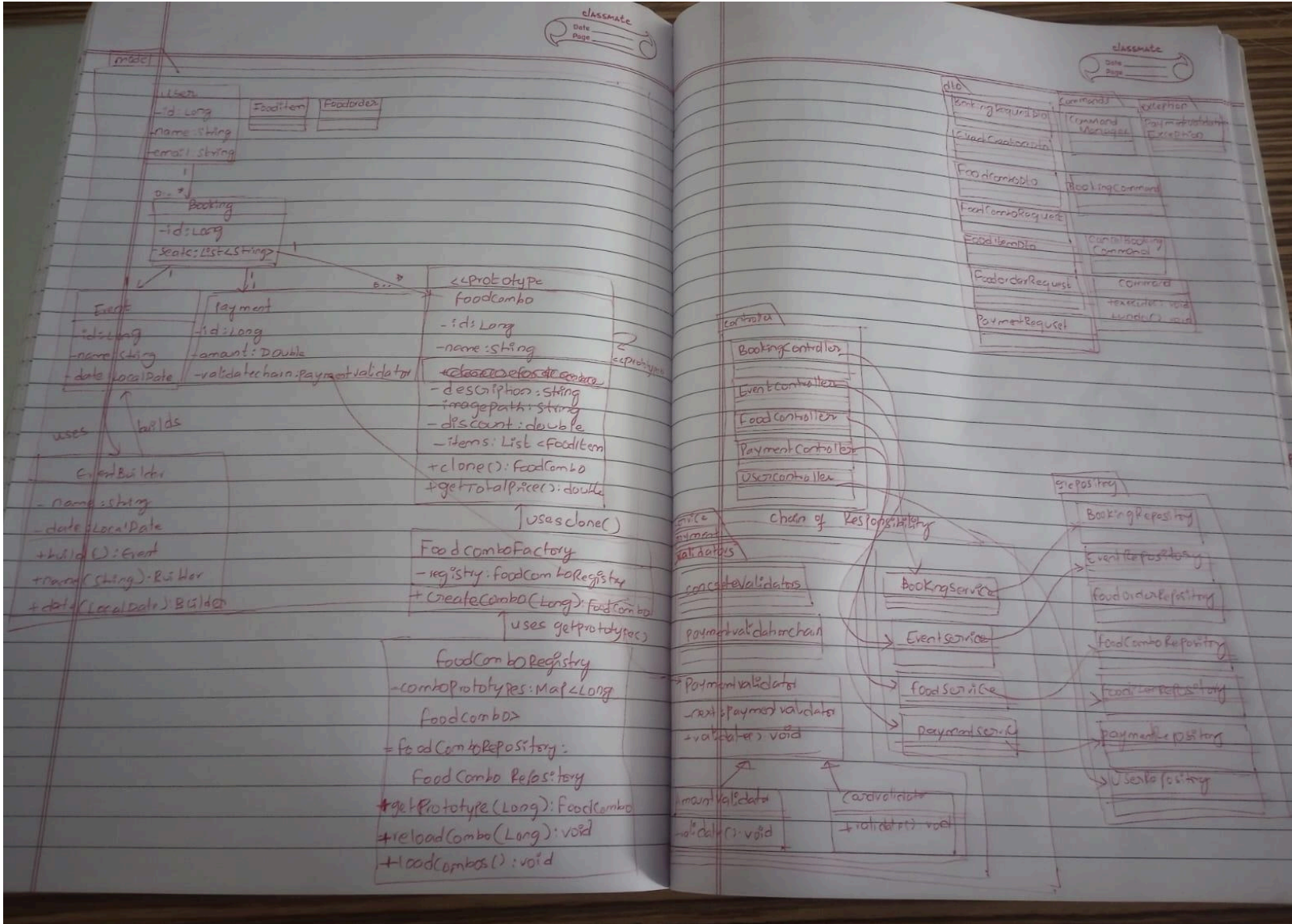
Use Case Diagram:



State Diagram:



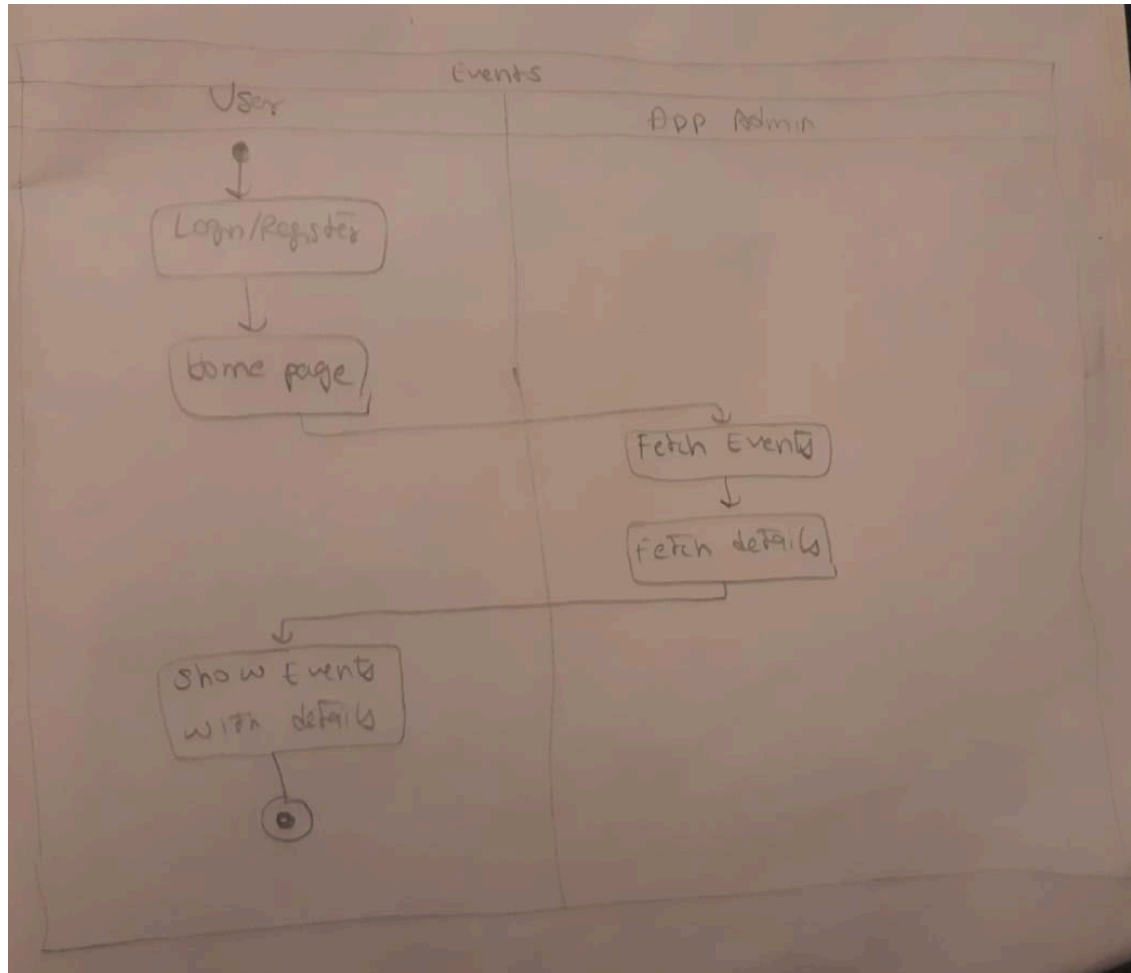
Class Diagram:



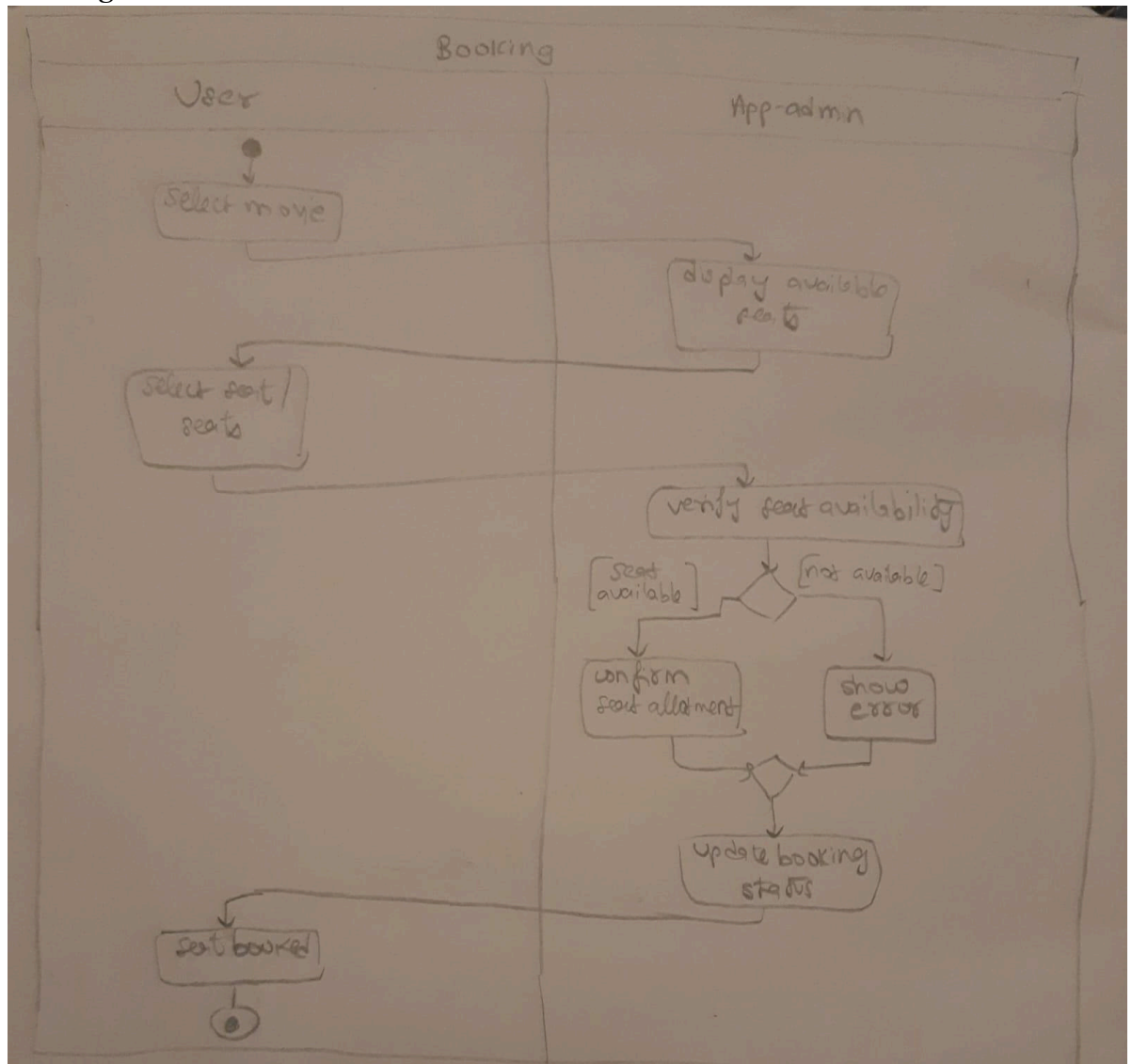
Activity Diagrams:

1. Major Use case

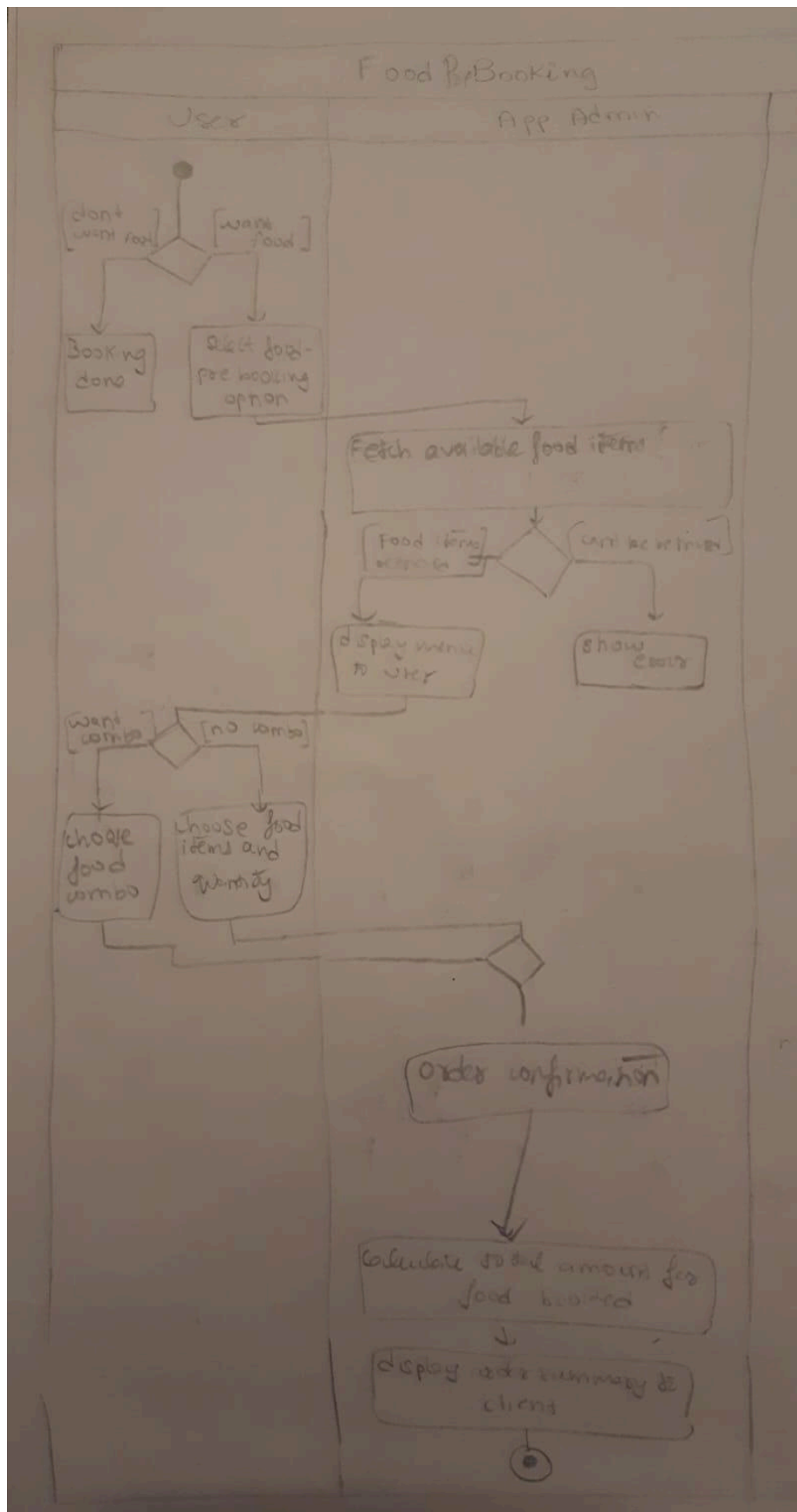
Displaying Events



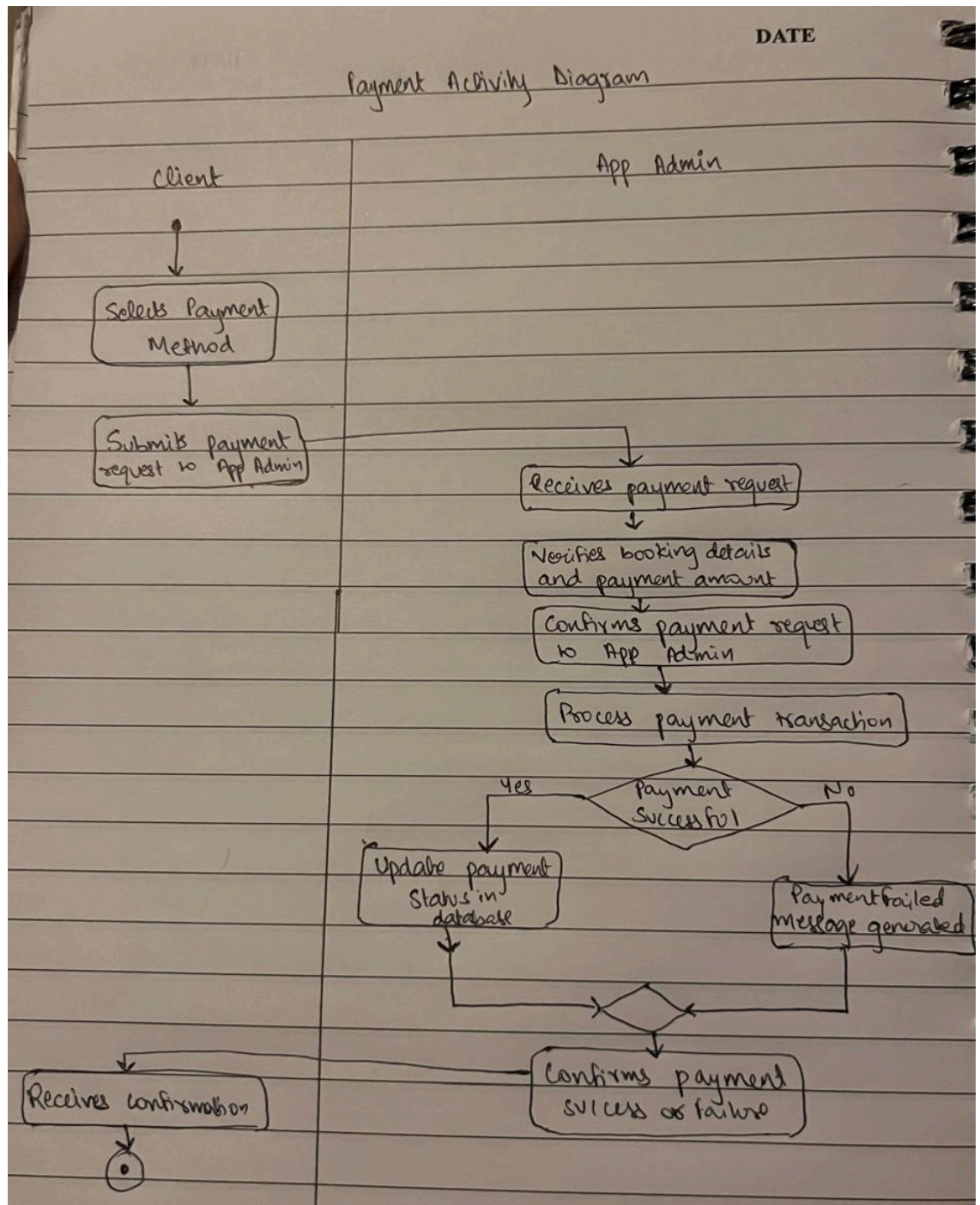
Booking



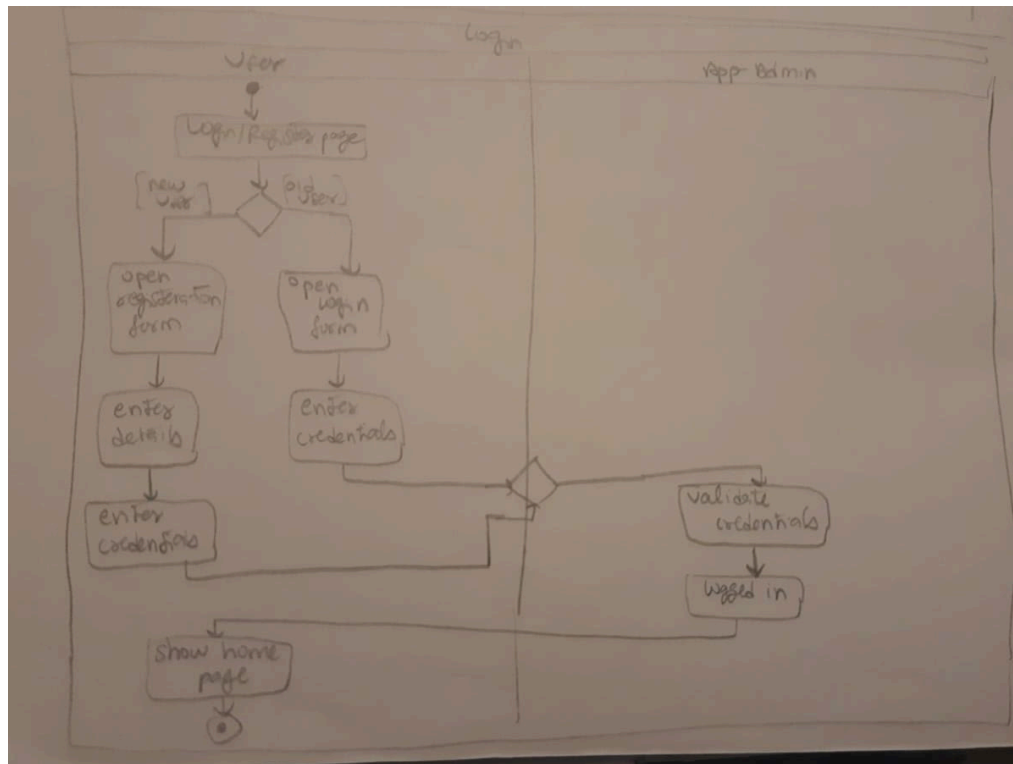
Preorder food



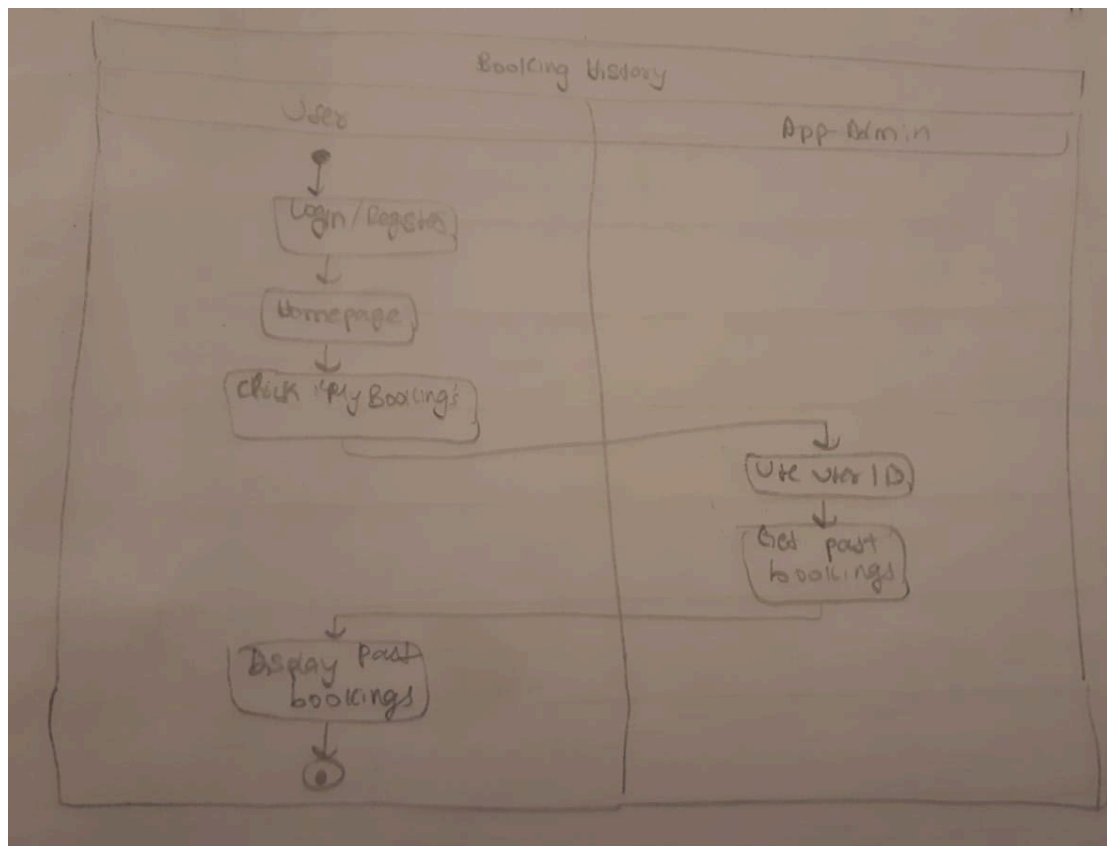
Payment



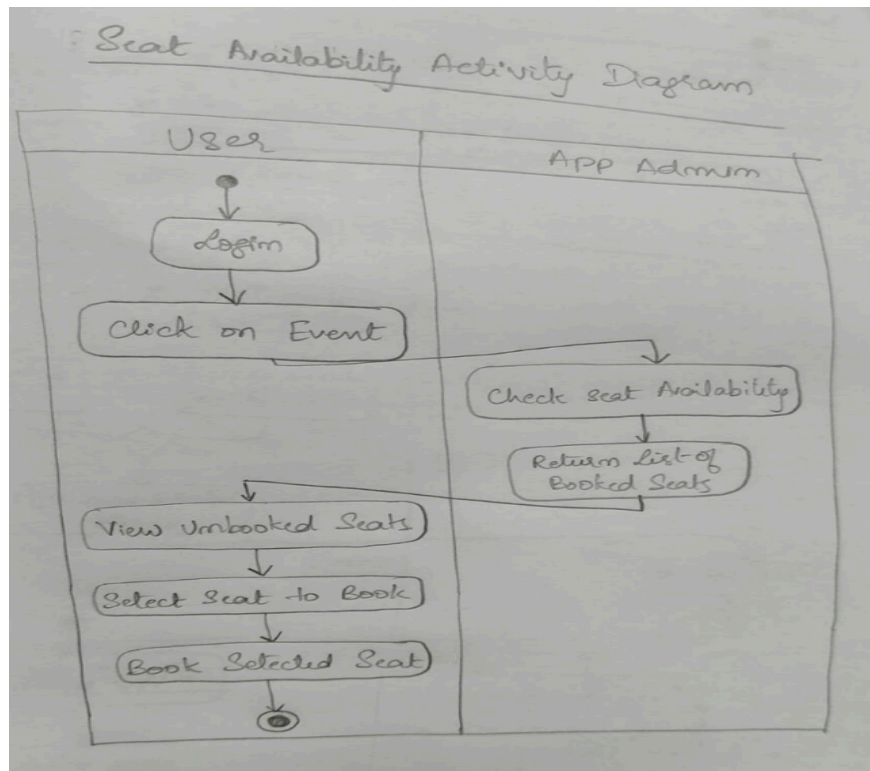
2. Minor Use case Login/Register



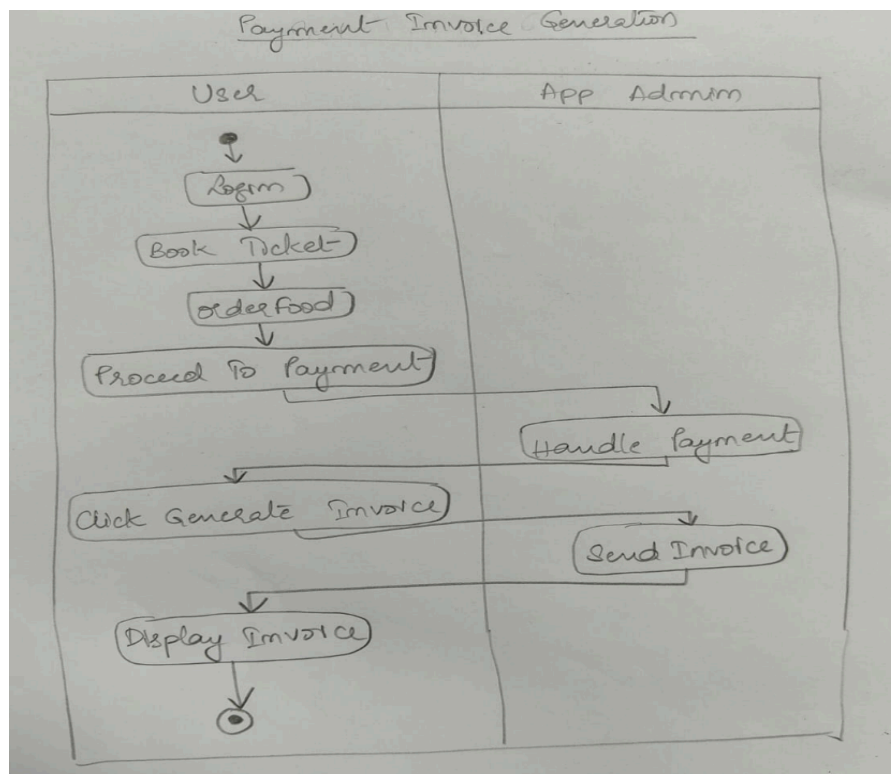
Booking History



Seat Availability



Payment Invoice Generation



Architecture Patterns, Design Principles, and Design Patterns:

Architecture Patterns

Model – View – Controller Pattern (MVC)

- **Model:** Represents the core data logic of the system. This includes entities such as User, Event, Seat, Booking, FoodCombo, and Payment. These models encapsulate database interactions and business rules.
- **View:** The user interface is built using HTML, CSS, and JavaScript. It displays event listings, seat layouts, food combos, payment gateways and booking summaries. It communicates with the backend via REST APIs to ensure a dynamic and responsive experience.
- **Controller:** Controllers handle incoming user requests, process input, and invoke appropriate service methods, they connect to services and repositories. Examples include EventController, BookingController, PaymentController etc.

Design Principles

- **Separation of Concerns:** Different modules handle logic, presentation, and data.
- **Open/Closed Principle:** The system supports extensibility without modifying existing code.
- **DRY (Don't Repeat Yourself):** Code reuse across services and controllers.
- **Low Coupling (Dependency Inversion Principle):** Via use of abstraction (abstract classes and methods).

Design Patterns

The Event Management System incorporates four key design patterns to enhance modularity and scalability.

- **Builder Pattern** simplifies the creation of complex Event objects by providing a fluent interface for setting properties, ensuring immutability and validation.
- **Chain of Responsibility Pattern:** For payment processing, decouples validation steps (e.g., amount checks, card/UPI verification) into linked handlers, enabling flexible and maintainable payment workflows.
- **Command Pattern:** encapsulates booking actions (e.g., create/cancel) as objects, supporting undo/redone functionality and decoupling UI interactions from backend logic.
- **Prototype Pattern:** optimizes food combo orders by allowing efficient cloning of predefined meal templates, reducing database load and ensuring consistency.

Github link to the Codebase:

https://github.com/kuku0809/Event_Management_System

Screenshots

UI:

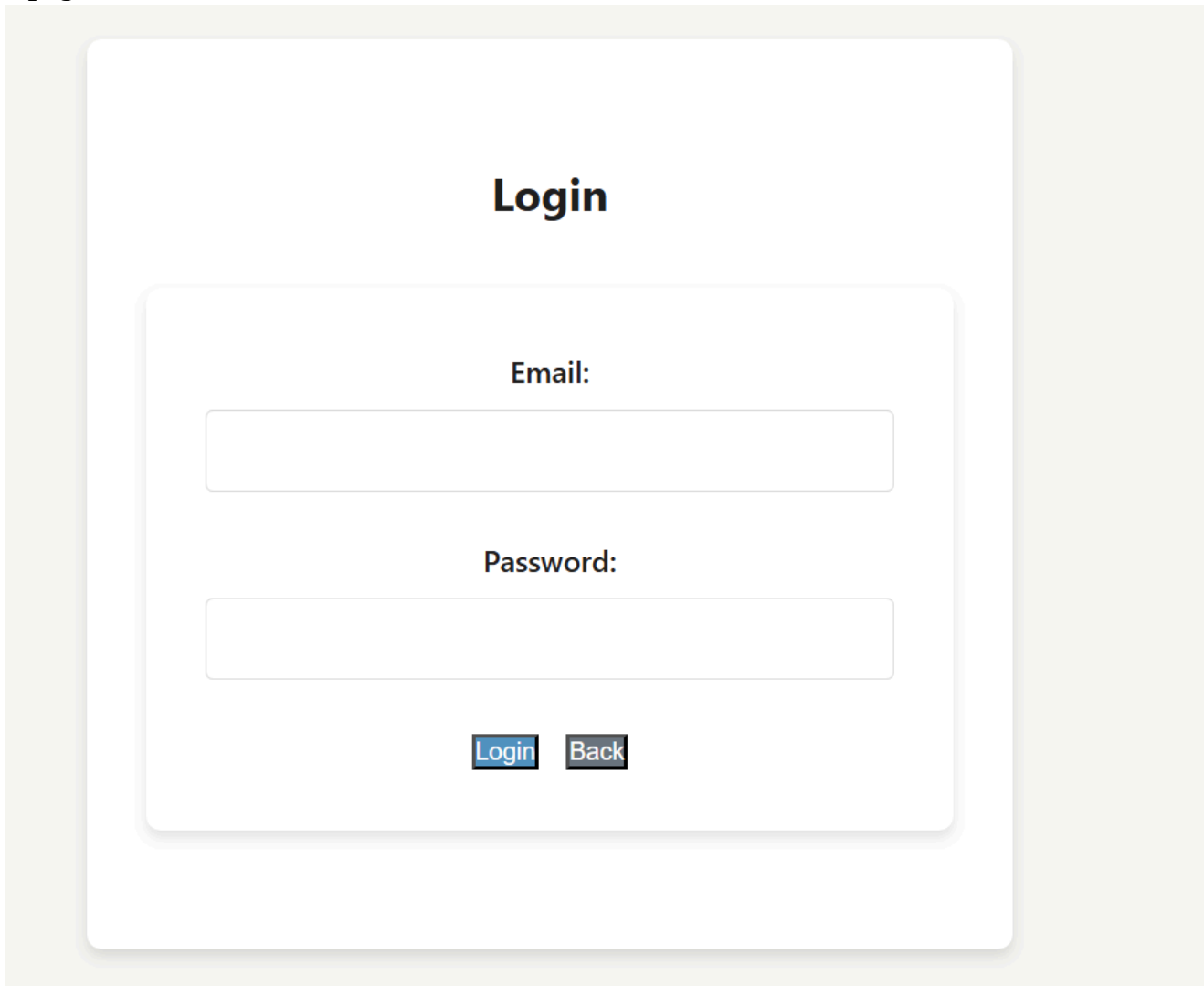
Login/Register Page:

Welcome to MoviesNMunchies

Login

Register

Login page:

The image shows a login page design. It features a light beige background. In the center is a white rounded rectangle with a subtle drop shadow. Inside this rectangle, the word "Login" is centered at the top in a bold, black, sans-serif font. Below "Login" is another white rounded rectangle, also with a drop shadow. Inside this inner rectangle, the label "Email:" is centered above a wide, light gray rectangular input field. Below the input field, the label "Password:" is centered above another wide, light gray rectangular input field. At the bottom of the inner rectangle, there are two buttons: a blue button with the text "Login" in white, and a black button with the text "Back" in white. Both buttons have a slight 3D effect with a shadow.

Login

Email:

Password:

Login Back

Home Page:

Book Tickets for Your Favorite Events

Discover and book tickets for concerts, movies, sports, and more!

[Explore Events](#)

Movie Display page:

Upcoming Events



Tangled

📍 Theater A

📅 4/10/2025

₹1500.00

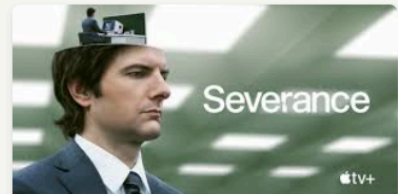
[Book Now](#)

Chennai Express

📍 Theater B

📅 4/15/2025

₹800.00

[Book Now](#)

Severance

📍 Theater C

📅 5/1/2025

₹2000.00

[Book Now](#)

Jumanji

📍 Theater D

📅 4/20/2025



SpyxFamily

📍 Theater A

📅 4/21/2025



Spiderman

📍 Theater D

📅 4/20/2025

Seat Selection page:

MoviesNMunchies

[Home](#)

[Events](#)

[My Bookings](#)

[Log](#)

Book Tickets

Event:

Tangled

1	2	3	4	5	6	7	8	9	10
11	12	13	14	15	16	17	18	19	20
21	22	23	24	25	26	27	28	29	30
31	32	33	34	35	36	37	38	39	40
41	42	43	44	45	46	47	48	49	50

No seats selected

Total Price:

[Book Now](#)

[Cancel](#)

Food booking page:

MoviesNMunchies

Home Events My Bookings Logout

Order Theatre Food

Individual Items Meal Combos



Popcorn
Large buttered popcorn
₹50.00
[Add to Cart](#)



Soda
32oz fountain drink
₹35.00
[Add to Cart](#)



Nachos
Cheesy nachos with jalapeños
₹70.00
[Add to Cart](#)



Candy
Assorted movie theater candy
₹40.00
[Add to Cart](#)



Fries
Salted french fires
₹80.00
[Add to Cart](#)



Pretzels
Salty doughy confectionary
₹90.00
[Add to Cart](#)

Your Cart

Total: ₹0.00

[Proceed to Payment](#) [Skip Food](#)

Food Combos:

Order Theatre Food

Individual Items **Meal Combos**



Family Combo

Great for sharing

Popcorn	₹50.00
Soda	₹35.00
Candy	₹40.00

~~₹125.00~~ **₹106.25** 15% OFF

Add Combo



Sweet Lovers

For those with a sweet tooth

Candy	₹40.00
Pretzels	₹90.00

~~₹130.00~~ **₹117.00** 10% OFF

Add Combo

Your Cart

Total: ₹0.00

[Proceed to Payment](#) [Skip Food](#)

Payment page:

Make Payment

Seats

₹1500.00

Total: ₹1500.00

Amount:

1500.00

Payment Method:

☒ Credit/Debit Card

☐ UPI

☐ Cash

Card Number:

1234 5678 9012 3456

Cardholder Name:

John Doe

Password:

Enter your card password

Complete Payment

Cancel

Booking History page:

MoviesNMunchies

Home

Events

My Bookings

Logout

My Bookings

Tangled

Theater A

4/10/2025

Seats: 1,2

Total: ₹3000.00

Booked on: 4/15/2025, 2:51:33 AM

Tangled

Theater A

4/10/2025

Seats: 3

Total: ₹1500.00

Booked on: 4/14/2025, 9:24:23 PM

Invoice:

MoviesNMunchies

Booking Invoice

Issued: 4/21/2025

Booking #22

Event Details

Description	Quantity	Unit Price	Total
Event Seats	2	₹1020.00	₹1020.00
Food & Beverages			
Fries	1	₹80.00	₹80.00
Subtotal			₹1100.00
Tax (if applicable)			₹0.00
Grand Total			₹1100.00

Thank you for your booking!

Individual contributions of the team members:

Name	Module worked on
Sinchana K C	Booking Flow,Seat Availability
Amitha Reddy	Event Management,Booking History
Kusuma Sanjaykumar	Food Utility,Login/Register
Maria George	Payment Gateway ,Invoice generation